

E-COMMERCE

Unit-3: Types of Electronic Payment Systems

Comprehensive Study Notes

1. Introduction to Electronic Payment Systems

1.1 Definition and Concept

Electronic Payment System (EPS) refers to any form of financial transaction that takes place electronically, rather than through physical cash or checks. It involves the transfer of funds from one party to another using digital technologies and networks.

Key Characteristics of Electronic Payment Systems:

Characteristic	Description
Speed	Transactions are processed in real-time or near real-time
Convenience	Payments can be made from anywhere, anytime
Security	Uses encryption and authentication technologies
Cost-Effective	Lower transaction costs compared to traditional methods
Traceability	Easy tracking and record-keeping of transactions

1.2 Evolution of Payment Systems

Ancient Times (Barter System)

- 600 BC (Coins & Currency)
- 17th Century (Paper Money & Cheques)
- 1950s (Credit Cards - Diners Club)
- 1970s (EFT - Electronic Funds Transfer)
- 1990s (Internet & Digital Payments)
- 2000s (Mobile Payments)
- 2010s (UPI, Contactless Payments)
- 2020s (CBDC, Blockchain, Tokenization)

1.3 Classification of Electronic Payment Systems

Type	Description	Examples
Digital Token-Based	Electronic tokens represent monetary value	E-cash, Bitcoin, Digital Gold
Credit Card-Based	Credit line extended by bank	Visa, Mastercard, American Express
Electronic Transfer	Direct bank-to-bank transfers	NEFT, RTGS, Wire Transfer
Mobile Payment	Payments via mobile devices	UPI, Apple Pay, Google Pay
Micro-Payment	Small transactions (< \$10)	Pay-Per-View, Music Downloads

2. Digital Token-Based Electronic Payment Systems

2.1 Definition of Digital Tokens

Digital Tokens (also called electronic tokens, digital cash, or e-cash) are electronic representations of monetary value that can be used for online transactions. They are created by a financial institution and secured through cryptographic techniques.

2.2 Characteristics of Digital Tokens

Characteristic	Description
Monetary Value	Backed by fiat currency (e.g., bank deposits)
Interoperability	Exchangeable with other payment systems
Security	Uses cryptography to prevent counterfeiting
Anonymity	Can be used without revealing identity
Divisibility	Can be divided into smaller units
Transferability	Can be transferred between parties

2.3 Types of Digital Tokens

A. Single-Purpose vs Multi-Purpose Tokens

Aspect	Single-Purpose Tokens	Multi-Purpose Tokens
Definition	Can only be used for one specific service or merchant	Accepted across multiple merchants and services
Flexibility	Limited - locked to one provider	High - usable anywhere
Examples	Gift cards, gaming tokens, Amazon Pay gift card	E-cash, Bitcoin, frequent flyer miles

B. On-Line vs Off-Line Tokens

Aspect	On-Line Tokens	Off-Line Tokens
Definition	Require real-time verification with bank	Can be used without immediate bank verification
Processing	Every transaction is validated online	Transactions are validated later (batch processing)
Security	Higher security - immediate fraud detection	Lower security - risk of double spending
Risk	Lower risk for merchant	Higher risk for merchant
Examples	Online banking, PayPal	Stored-value cards, Prepaid phone cards

C. Anonymous vs Identifiable Tokens

Aspect	Anonymous Tokens	Identifiable Tokens
Privacy	Complete privacy - no identity linked	Identity linked to transaction
Traceability	Cannot be traced	Full traceability
Use Case	Privacy-sensitive purchases	Tax compliance, anti-money laundering
Risk	Potential for illegal use	Privacy concerns
Examples	Digital cash, Cryptocurrency	Credit cards, Bank transfers

2.4 E-Cash (Digital Cash)

Definition: E-cash is a digital form of cash that uses cryptographic techniques to create anonymous, untraceable electronic money.

Key Features:

- Created by banks
- Stored on user's computer or smart card
- Can be spent anonymously
- Cannot be counterfeited (cryptographic protection)

Working of E-Cash:

1. User withdraws e-cash from bank (deducted from account)
2. Bank issues digitally signed electronic coins
3. User sends e-cash to merchant for payment
4. Merchant verifies the digital signature with the bank
5. Bank confirms coins are valid and not double-spent
6. Merchant deposits the e-cash with the bank

Advantages:

- Anonymity for users
- Low transaction costs
- Instant settlement
- No need for credit history

Disadvantages:

- Risk of theft (if not secured)
- Not widely accepted
- Storage issues (hard disk crash)
- Tax evasion concerns

2.5 Cryptocurrency (Bitcoin)

Definition: Cryptocurrency is a decentralized digital currency that uses blockchain technology and cryptography for security. Bitcoin, created in 2009 by Satoshi Nakamoto, is the first and most well-known cryptocurrency.

Key Features:

- Decentralized (no central authority)
- Limited supply (Bitcoin has 21 million cap)
- Pseudonymous (not fully anonymous)
- Global accessibility
- Volatile value

Working of Blockchain-Based Payments:

1. Transaction initiated and broadcast to network
2. Mining nodes validate the transaction

3. Validated transactions added to a block
4. Block is chained to previous blocks (Blockchain)
5. Transaction is confirmed (irreversible)
6. Recipient receives the cryptocurrency

Advantages:

- No intermediaries
- Low international transaction fees
- Fast cross-border transfers
- Transparent and secure

Disadvantages:

- High volatility
- Regulatory uncertainty
- Energy consumption (mining)
- Limited acceptance

3. Credit Card-Based Electronic Payment Systems

3.1 Introduction to Credit Cards

Definition: A credit card is a payment card issued by a financial institution that allows cardholders to borrow funds to make purchases, with the agreement to pay back the borrowed amount plus interest if not paid in full by the due date.

3.2 Key Parties in Credit Card Transactions

Party	Role
Cardholder	The consumer making the purchase
Merchant	The business accepting the card
Issuing Bank	Bank that issued the card to the cardholder
Acquiring Bank	Bank that processes payments for the merchant
Card Network	Visa, Mastercard, American Express - facilitates communication
Payment Gateway	Technology that transmits transaction data

3.3 Credit Card Transaction Process

1. Cardholder presents card to merchant (online or in-store)
2. Merchant sends transaction to acquiring bank via payment gateway
3. Acquiring bank forwards to card network (Visa/Mastercard)
4. Card network routes to issuing bank for authorization
5. Issuing bank approves/declines based on credit limit & fraud checks
6. Authorization response flows back through network → acquirer → merchant
7. Merchant completes the sale
8. Settlement: Issuing bank transfers funds to acquiring bank
9. Acquiring bank deposits funds into merchant's account
10. Cardholder receives monthly statement and pays issuing bank

3.4 SET Protocol (Secure Electronic Transaction)

SET is a protocol developed by Visa and Mastercard to ensure secure credit card transactions over the internet. It provides confidentiality, integrity, and authentication using digital certificates.

Key Features of SET:

- Encrypts credit card details and order information separately
- Uses dual signatures to link order and payment info
- Digital certificates verify all parties (cardholder, merchant, gateway)
- Merchant never sees the cardholder's credit card number

SET Transaction Flow:

1. Cardholder registers and obtains digital certificate
2. Merchant obtains digital certificate from acquirer
3. Payment gateway obtains digital certificate
4. Purchase request sent with dual signature (order info + payment info)
5. Merchant verifies cardholder certificate
6. Merchant forwards payment info to gateway
7. Gateway verifies and processes payment
8. Gateway sends confirmation to merchant
9. Merchant sends confirmation to cardholder

3.5 Credit Card Fraud and Security

Common Fraud Types:

Fraud Type	Description	Prevention
Card Skimming	Copying magnetic stripe data	EMV chips
Phishing	Fake emails/sites to steal details	Two-factor authentication
Card Not Present (CNP)	Online transactions without physical card	CVV, 3D Secure
Identity Theft	Using stolen personal information	Biometric verification
Counterfeit Cards	Creating fake cards	EMV chips

Security Measures:

1. **CVV (Card Verification Value):** 3-digit code on back of card
2. **3D Secure:** Verified by Visa, Mastercard SecureCode
3. **Tokenization:** Replacing card number with unique token
4. **Two-Factor Authentication:** Additional verification step
5. **Transaction Monitoring:** Real-time fraud detection

EMV Chip Technology:

EMV (Europay, Mastercard, Visa) chips generate a unique cryptogram for each transaction, making them far more secure than magnetic stripe cards. The chip creates a one-time code that cannot be reused if intercepted.

4. Electronic Fund Transfer Systems

4.1 NEFT (National Electronic Funds Transfer)

Feature	Details
Full Form	National Electronic Funds Transfer
Operated By	Reserve Bank of India (RBI)
Mode	Batch processing (half-hourly settlement)
Availability	24×7 (round the clock)
Transfer Limit	No minimum; maximum varies by bank
Settlement Time	Up to 2 hours (within batches)
Best For	Small to medium value transactions

4.2 RTGS (Real Time Gross Settlement)

Feature	Details
Full Form	Real Time Gross Settlement
Operated By	Reserve Bank of India (RBI)
Mode	Real-time (continuous settlement)
Availability	24×7 (round the clock)
Transfer Limit	Minimum '2 lakh; no maximum
Settlement Time	Immediate (real-time)
Best For	High-value urgent transactions

4.3 Wire Transfer

Wire Transfer is an electronic transfer of funds between banks or financial institutions via networks such as SWIFT (Society for Worldwide Interbank Financial Telecommunication). It is commonly used for international transfers.

Key Features:

- Used for domestic and international transfers
- SWIFT network facilitates international wire transfers
- Higher fees compared to NEFT/RTGS
- Settlement can take 1-5 business days
- Requires sender and recipient bank details

Aspect	NEFT	RTGS	Wire Transfer
Settlement	Batch (half-hourly)	Real-time	1-5 days
Minimum Amount	No minimum	₹2 lakh	Varies
Speed	Moderate	Immediate	Slow
Best Use	Routine small transfers	Large urgent payments	International transfers

5. Mobile Payment Systems

5.1 UPI (Unified Payments Interface)

Definition: UPI is a real-time payment system developed by the National Payments Corporation of India (NPCI) that enables instant money transfer between bank accounts using a mobile platform.

How UPI Works:

1. User downloads a UPI-enabled app (Google Pay, Paytm, PhonePe)
2. User registers with mobile number and creates a UPI PIN
3. A unique UPI ID (e.g., name@bank) is created
4. To pay: Enter recipient's UPI ID/scan QR code, enter amount & UPI PIN
5. Payment is processed instantly via the Immediate Payment Service (IMPS)
6. Both sender and receiver get confirmation in real-time

Key Features of UPI:

- **Single App:** Access multiple bank accounts from one app
- **24×7:** Available round the clock
- **Instant:** Real-time fund transfer
- **Interoperable:** Works across all banks
- **Low Cost:** Minimal or no transaction charges
- **Secure:** Two-factor authentication (MPIN + device binding)

5.2 Popular UPI Apps

Google Pay (GPay):

- Developed by Google
- Uses UPI for instant transfers
- Features: Bill payments, merchant payments, rewards
- Security: Device binding + UPI PIN

Paytm:

- Developed by One97 Communications
- Full digital wallet + UPI integration
- Features: Mobile recharge, bill pay, travel, shopping
- Security: PIN, fingerprint, device lock

PhonePe:

- Developed by Flipkart (Walmart)
- UPI-based payment platform
- Features: Recharges, bill payments, insurance, investments
- Security: UPI PIN + device authentication

5.3 NFC Payments (Near Field Communication)

NFC is a short-range wireless technology (operating at 13.56 MHz, range < 10 cm) that enables two devices to communicate when tapped or brought close together. Used for contactless payments.

How NFC Payments Work:

1. User adds card to digital wallet (Apple Pay, Google Wallet)
2. At checkout, user taps phone near NFC-enabled POS terminal
3. Phone and terminal establish secure NFC connection
4. Transaction token (not actual card number) is transmitted
5. Biometric/PIN verification on phone
6. Payment authorized & confirmed

NFC Payment Platforms:

Platform	Developer	Key Features
Apple Pay	Apple Inc.	Works on iPhone, Apple Watch; Face ID/Touch ID; tokenization
Google Wallet	Google	Works on Android; virtual card numbers; Passes integration
Samsung Pay	Samsung	MST + NFC; works with older terminals

Advantages of NFC Payments:

- Fast - tap and go in seconds

- Secure - tokenization and biometric authentication
- Convenient - no need to carry physical wallet
- Hygienic - contactless, no touching terminals

6. Security in Electronic Payment Systems

6.1 Encryption

Definition: Encryption is the process of converting plain text data into ciphertext using an algorithm and a key, making it unreadable to unauthorized parties.

Types of Encryption:

Type	Description	Example
Symmetric Encryption	Same key for encryption and decryption	AES, DES, 3DES
Asymmetric Encryption	Public key for encryption, private key for decryption	RSA, ECC, DSA

6.2 SSL/TLS (Secure Sockets Layer / Transport Layer Security)

SSL/TLS are cryptographic protocols that provide secure communication over a computer network. TLS is the successor to SSL and is widely used for securing web traffic (HTTPS).

How SSL/TLS Works:

1. Client connects to server requesting secure connection
2. Server sends its digital certificate (containing public key)
3. Client verifies the certificate with a Certificate Authority (CA)
4. Client generates a session key, encrypts with server's public key
5. Server decrypts session key with its private key
6. Both parties now communicate using symmetric encryption (session key)

6.3 Digital Signatures

Definition: A digital signature is a cryptographic technique that verifies the authenticity and integrity of a digital message or document. It provides non-repudiation, meaning the sender cannot deny having sent the message.

Working:

1. Sender creates a hash of the message
2. Hash is encrypted with sender's private key (digital signature)
3. Message + digital signature sent to receiver
4. Receiver decrypts signature using sender's public key
5. Receiver computes hash of received message
6. If both hashes match, signature is verified

6.4 Digital Certificates

Definition: A digital certificate is an electronic document issued by a Certificate Authority (CA) that binds a public key to an identity (person, organization, or device). It follows the X.509 standard.

Contents of a Digital Certificate:

- Subject (owner) name and organization
- Subject's public key
- Certificate Authority (issuer) name
- Serial number
- Validity period (start and end dates)
- Digital signature of the CA

6.5 Authentication Methods

Two-Factor Authentication (2FA):

2FA requires two different types of authentication factors before granting access:

Factor Type	Examples
Something you KNOW	Password, PIN, security question
Something you HAVE	OTP on phone, hardware token, smart card
Something you ARE	Biometric (fingerprint, face, iris)

Biometric Authentication:

- **Fingerprint Scanning:** Fingerprint readers on phones/laptops
- **Facial Recognition:** Face ID, Windows Hello

- **Iris Scanning:** High-accuracy eye pattern recognition
- **Voice Recognition:** Voiceprint for phone banking
- **Behavioral Biometrics:** Typing patterns, mouse movements

6.6 Security Measures Summary

Security Measure	Description
Encryption	Data encrypted during transmission (SSL/TLS)
Authentication	Verifying identity (PIN, OTP, Biometric)
Tokenization	Replace sensitive data with tokens
PCI DSS Compliance	Payment Card Industry Data Security Standard
Firewalls	Protect network from unauthorized access
Intrusion Detection	Monitor for suspicious activity
Regular Audits	Security assessments and compliance checks
Employee Training	Awareness of security threats and best practices

7. Risks in Electronic Payment Systems

7.1 Types of Risks

Electronic Payment Systems Risks

- Transaction Risk (chargeback, authorization, settlement)
- Credit Risk (cardholder default, merchant insolvency)
- Operational Risk (tech failures, human errors)
- Fraud Risk (CNP fraud, phishing, identity theft)

7.2 Fraud Prevention Measures

Measure	Description
Address Verification System (AVS)	Verify billing address matches cardholder record
CVV Verification	Require 3-digit CVV code
3D Secure	Additional authentication via OTP
Risk Scoring	Rate transactions for fraud probability
Velocity Checks	Monitor transaction frequency from same card/IP
Machine Learning	AI models detect unusual patterns
Device Fingerprinting	Identify suspicious devices
Geolocation Verification	Check location consistency with billing address

8. Comparative Analysis

8.1 Comparison of Payment Systems

Feature	Digital Cash	Smart Card	Credit Card	UPI/Mobile
Anonymity	High	Low	Low	Medium
Transaction Cost	Low	Medium	High	Low
Speed	High	High	High	Instant
Security	Medium	High	Medium	High
Acceptance	Low	High	Very High	High (India)
Infrastructure	Low	Medium	High	Low
Traceability	Low	High	High	High
User Convenience	Medium	High	Very High	Very High

8.2 Suitability Matrix

Payment System	Best For	Examples
Digital Cash	Small transactions, privacy-sensitive	Micro-payments, Gig economy
Smart Cards	Multiple applications, offline use	Public transport, Student IDs
Credit Cards	Online purchases, consumer convenience	E-commerce websites, Retail
Debit Cards	Daily purchases, budget-conscious	Grocery, Fuel, Utility bills
UPI/Mobile	Instant transfers, peer payments	Bill splitting, Merchant payments
Cryptocurrency	Cross-border, high-value	International remittance

Questions

Explain digital token-based electronic payment systems with examples. (10 marks)

What are smart cards? Discuss their role in electronic payment systems. (10 marks)

Explain the credit card-based electronic payment system with a diagram. (10 marks)

Discuss the various risks associated with electronic payment systems and their mitigation. (10 marks)

What is EDI? Explain its applications in business with examples. (10 marks)

Define electronic payment system. Discuss its key characteristics and evolution. (5 marks)

Explain the classification of electronic payment systems with examples. (5 marks)

Write a short note on UPI and how it works. (5 marks)

Distinguish between NEFT and RTGS. (5 marks)

Explain the role of encryption, SSL/TLS, and digital signatures in payment security. (5 marks)

What is SET protocol? Explain its working. (5 marks)

Compare any four types of electronic payment systems. (5 marks)

— End of Unit 3: Types of Electronic Payment Systems —